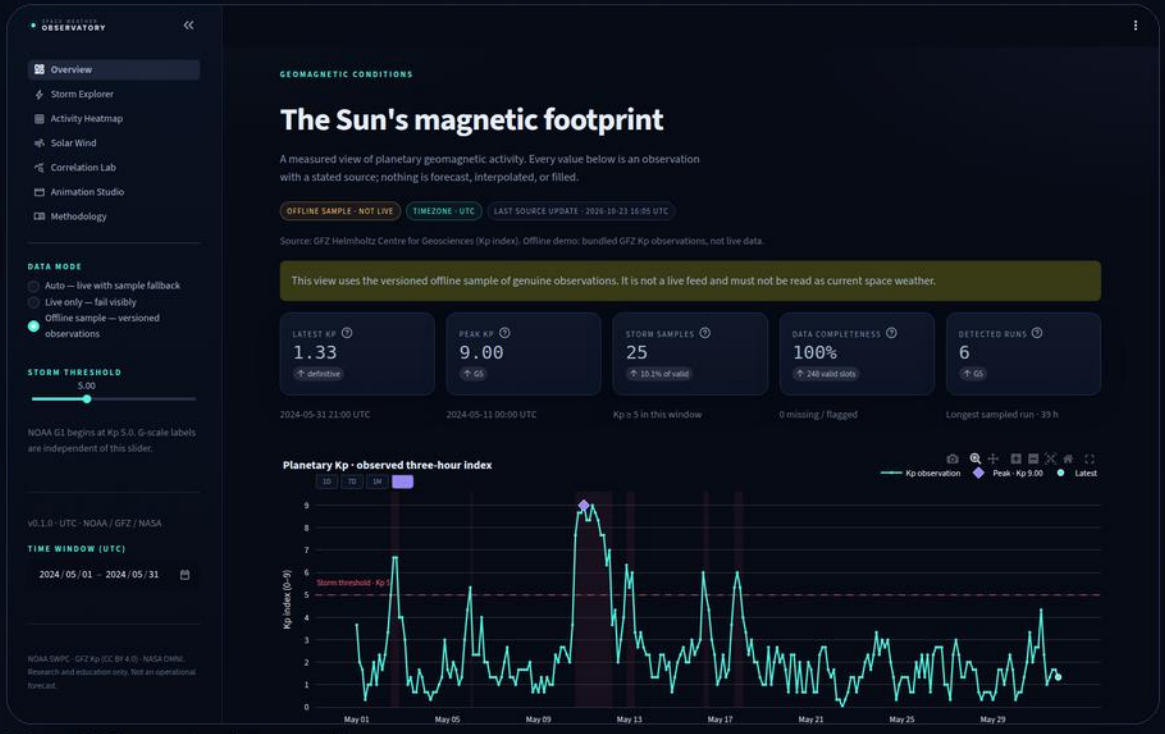
WHEN THE SUN DISRUPTS EARTH

When the Sun disrupts Earth.

A clear, honest look at geomagnetic storms: real data, visible sources, and no made-up values.

NOAA / GFZ / NASA · 58 TESTS · FREE · OPEN SOURCE



May 2024 sample data · not live

github.com/Shahbaz2104/space-weather-observatory

Three ways a chart can lie.

A chart can mislead without a single wrong number.

01 **A gap drawn as calm**

Missing readings become a flat line.

02 **A smooth line drawn as real data**

Smoothing invents values that were never measured.

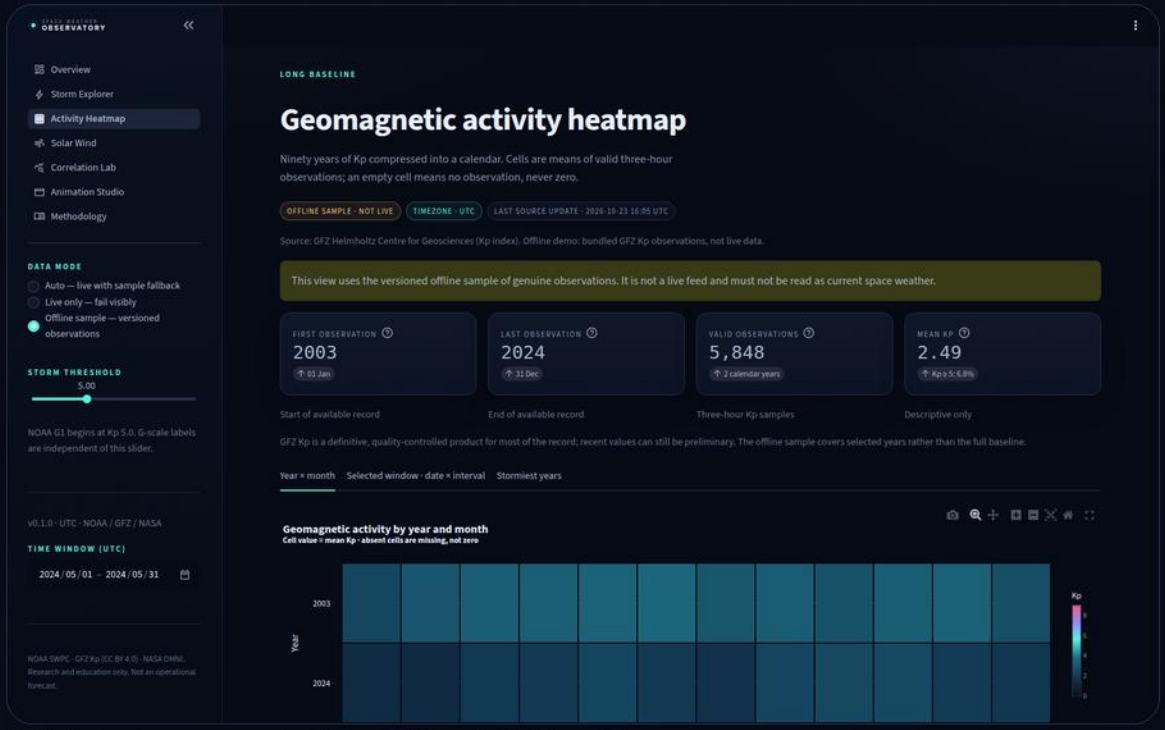
03 **One high number mistaken for a storm**

A single reading is not a full storm.

Seven linked views. One shared timeline.

From the latest Kp reading to a long-term heatmap, every page uses UTC and keeps gaps visible.

Overview · Storm Explorer · Activity Heatmap · Solar Wind · Correlation Lab · Animation Studio · Methodology



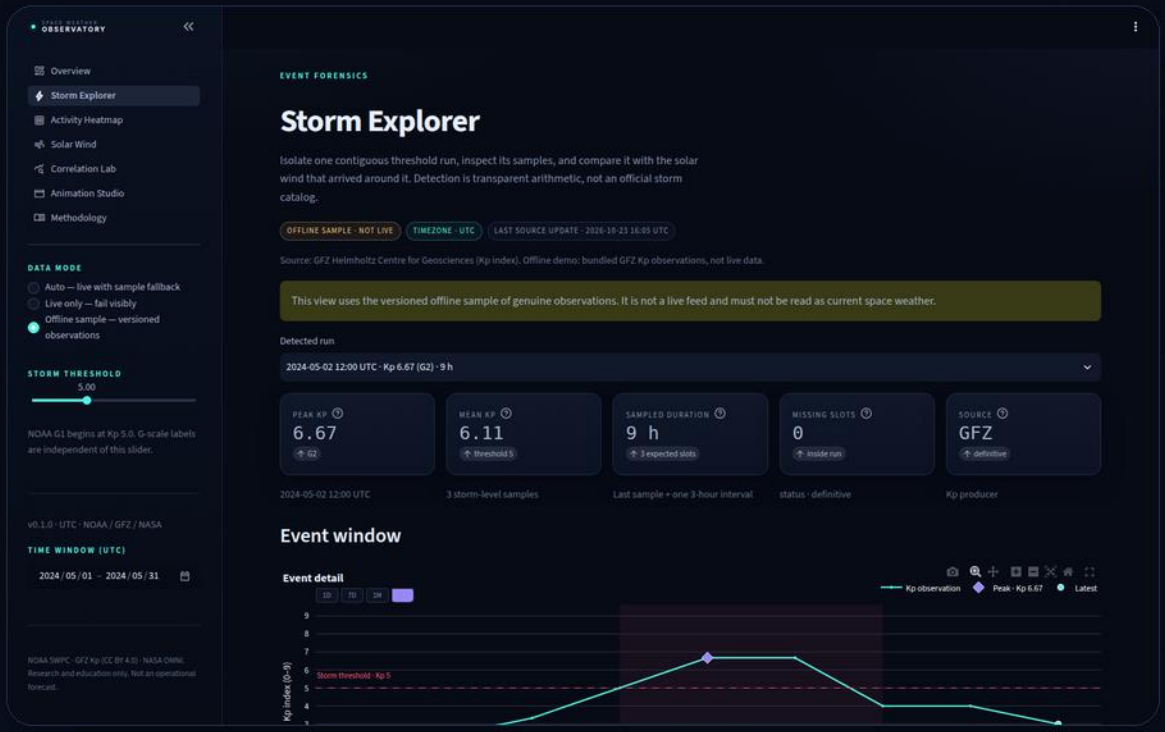
Activity heatmap · sample data, 2003 and 2024 · not live

github.com/Shahbaz2104/space-weather-observatory

Missing stays missing.

No filled gaps. No fake zeros. No hidden gaps.

Charts keep the edges of every gap, so a long record cannot be made to look complete.



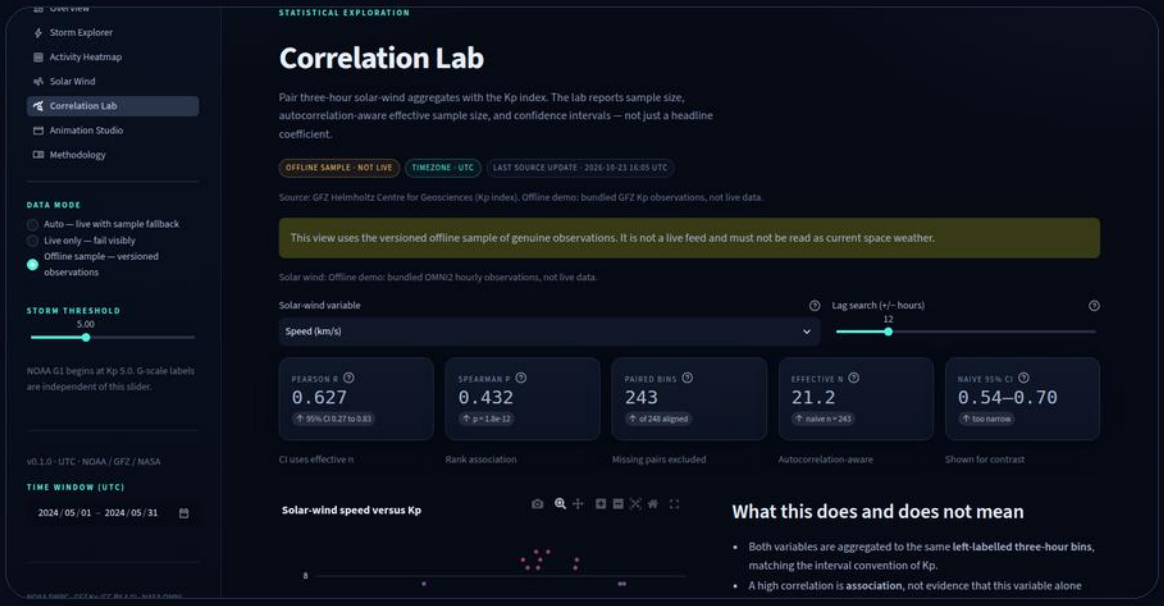
Storm Explorer · storm length and missing readings · sample, not live

Every download can include a source record: where the data came from, when, the units, settings, and warnings.

Moving together is not proof.

The lab shows how much independent evidence is behind the headline number.

CORRELATION	REAL EVIDENCE	READINGS
0.627	21.2	243
1 means a perfect match	not 243 separate readings	three hours each



May 2024 sample · not live · solar wind speed and Kp

CHECK THE EVIDENCE

Seven pages. Real data.

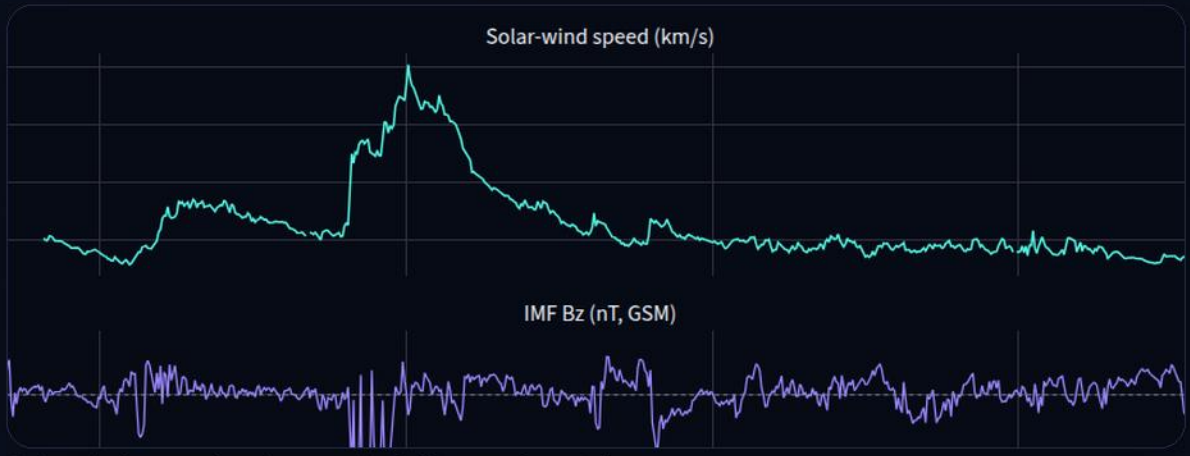
Built with Python and Streamlit. Every page shows the limits as clearly as the numbers.

LIVE DEMO
https://space-weather-observatory-4xedxr5f2n5vpym7t5ix7p.streamlit.app

REPOSITORY
github.com/Shahbaz2104/space-weather-observatory

Live data and sample data are labelled. Not a forecast service.

58 TESTS | AUTOMATED CHECKS | OPEN SOURCE



Solar wind · speed and magnetic field · sample, not live